

Institute of Technology of Cambodia

Department of Information and Communication



Internet Programming

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Topic: Project Proposal

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1. Executive Summary

The **Recipe Sharing Web Application** (*CookShare*) is a full-stack web platform designed to facilitate culinary knowledge exchange among home cooks, culinary enthusiasts, and food creators. The platform allows users to create, document, publish, and explore structured recipes featuring comprehensive ingredient breakdowns, preparation times, difficulty levels, and step-by-step instructions.

To maintain high community standards and ensure content safety, the platform incorporates a role-based moderation system where users can flag inappropriate or hazardous submissions, and moderators can review and hide reported recipes.

The application is engineered strictly adhering to the requirements of the *Internet Programming I* course at the Institute of Technology of Cambodia (ITC), built with a decoupled architecture utilizing **Vue 3 with TypeScript, Vite, and Pinia** on the frontend, and **NestJS 11 with Prisma ORM and PostgreSQL** on the backend.

2. Purpose and Scope

2.1 Problem Statement

Culinary hobbyists frequently encounter disorganized cooking blogs cluttered with advertisements, lengthy personal anecdotes, unstructured ingredient lists, and unverified content. Furthermore, open communities frequently struggle with spam, off-topic posts, or unsafe cooking instructions. There is a need for a clean, structured, community-driven recipe application that emphasizes standardized recipe formatting, swift ingredient-based querying, and community moderation.

2.2 Proposed Solution

The *CookShare* web application provides:

- A standardized recipe composition interface (title, description, prep/cook time, difficulty, category, ingredients with units, and step-by-step instructions).
- Fast keyword search, category filtering, and paginated browsing.

- Secure, token-based authentication (JWT) with password hashing (bcrypt).
- A robust moderation pipeline: members can report problematic recipes with specific rationale, and moderators have dedicated controls to inspect reports and hide/unhide posts.
- A fully responsive, accessible Single Page Application (SPA) communicating over a typed RESTful API.

2.3 Target Audience

Home Cooks & Food Enthusiasts (Member): Individuals seeking to document their favorite dishes, discover new meal ideas, and share cooking techniques.

Administrators: Community curators responsible for inspecting reported content, enforcing guidelines, and keeping the platform clean.

2.4 Constraints, Assumptions, and Dependencies

Architectural Constraints:

- Frontend must be built with Vue 3, TypeScript, Vite, and Pinia.
- Backend must be built using NestJS 11 with TypeScript.
- Database PostgreSQL managed through Prisma ORM.

Dependencies:

- Node.js runtime (v20+ LTS recommended).
- PostgreSQL 15+ database instance.
- Automated testing tools: Jest, Supertest, and Vitest.

3. System Architecture & Technology Stack

The system follows a modern decoupled Client-Server architecture:

- **Client Tier:** Vue 3 Single Page Application (SPA) served via Vite. Handles UI rendering, local client-side validation, routing guards, and state management.
- **Server Tier:** NestJS 11 modular REST API. Implements business logic, request validation pipes, authentication guards, and role-based authorization.
- **Data Tier:** PostgreSQL relational database accessed exclusively through the Prisma ORM query engine and migration toolkit.

4. Data Management & Entity Design

4.1 Data Entities & Relationships

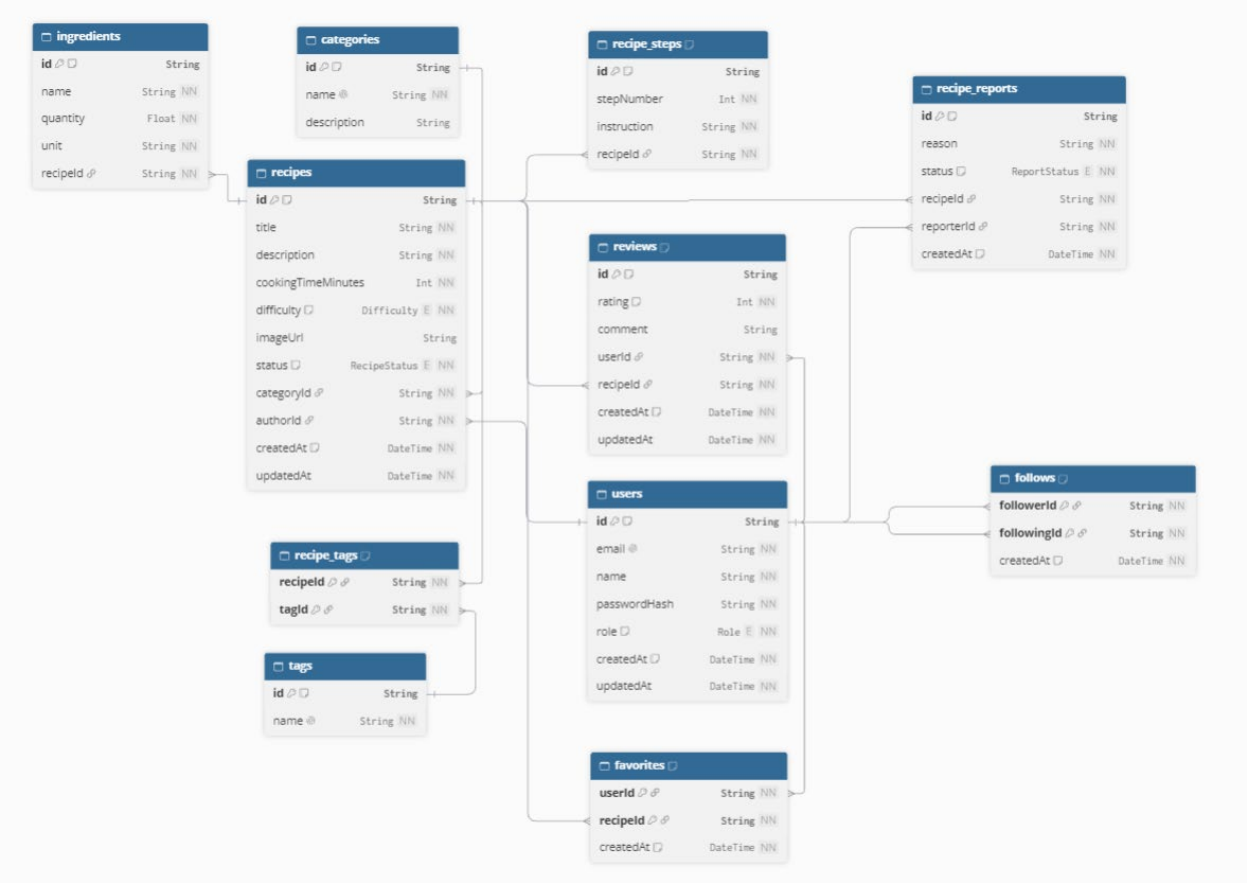
The application defines four core related entities satisfying and exceeding the course requirement of at least 3 related entities with 1:N relations:

1. **User: Represents registered members and moderators.**
    - 1:N with Recipe (Author)
    - 1:N with RecipeReport (Reporter)
    - 1:N with Review (Reviewer)
    - 1:N with Favorite (Saved recipes)
    - 1:N Self-Referencing with Follow (as follower and following)
2. **Recipe: Central entity of the platform.**
    - N:1 with User (Author)
    - N:1 with Category (Normalized classification)
    - 1:N with Ingredient (Itemized ingredients)
    - 1:N with RecipeStep (Ordered step-by-step preparation)
    - 1:N with RecipeTag (Many-to-Many junction with Tag)
    - 1:N with Review (Feedback and ratings)
    - 1:N with Favorite (Bookmark count)
    - 1:N with RecipeReport (Moderation flags)
3. **Category: Formal recipe classification (e.g., "Soup", "Dessert", "Khmer Cuisine").**
    - 1:N with Recipe.
4. **Ingredient: Structured quantity, unit, and item linked to a recipe.**
    - N:1 with Recipe.
5. **RecipeStep: Sequenced instructions with ordered position numbers.**
    - N:1 with Recipe.
6. **Tag & RecipeTag: Cross-cutting dietary and lifestyle labels (e.g., "vegan", "spicy", "quick").**
    - M:N between Recipe and Tag via junction table RecipeTag.
7. **Review: Ratings (1–5) and optional text comments.**
    - N:1 with User and N:1 with Recipe (Unique compound constraint: 1 review per user per recipe).
8. **Favorite: Direct bookmarking junction.**

- N:1 with User and N:1 with Recipe (Compound primary key).

9. **Follow: Social graph tracking follower-creator connections.**
    - N:1 with User (Follower) and N:1 with User (Following).
10. **RecipeReport: Content moderation tickets for inappropriate submissions.**
    - N:1 with Recipe and N:1 with User.

4.2 Database Schema



Reference: https://dbdiagram.io/d/RecipeSharing-6a13bd3eb62396d22c5d57c1

5. Functional Requirements & User Stories

Feature & Action	Guest	Member	Admin
Browse & Search Published Recipes	Allowed	Allowed	Allowed
Filter by Category & Tags	Allowed	Allowed	Allowed
View Recipe Details, Ingredients & Steps	Allowed	Allowed	Allowed
View Public Reviews & Ratings	Allowed	Allowed	Allowed
Register & Login	Allowed	N/A	N/A
Create New Recipe (with Steps & Ingredients)	Forbidden	Allowed	Allowed
Edit / Delete Own Recipe	Forbidden	Allowed	Allowed
Edit / Delete Other User's Recipe	Forbidden	Forbidden	Allowed
Assign Existing Tags & Category to Recipe	Forbidden	Allowed	Allowed
Create / Manage Master Categories & Tags	Forbidden	Forbidden	Allowed
Submit / Update Rating & Review (1 per recipe)	Forbidden	Allowed	Allowed
Delete Any User's Inappropriate Review	Forbidden	Forbidden	Allowed
Bookmark Recipe to Favorites (Add / Remove)	Forbidden	Allowed	Allowed
Follow / Unfollow Other Creators	Forbidden	Allowed	Allowed
Submit Inappropriate Recipe Report	Forbidden	Allowed	Allowed
View Moderation Report Queue	Forbidden	Forbidden	Allowed
Hide / Restore Reported Recipe	Forbidden	Forbidden	Allowed
Dismiss / Resolve Content Reports	Forbidden	Forbidden	Allowed

6. Conclusion & Next Steps

This project proposal establishes a complete, robust, and verifiable software requirements specification for the **Recipe Sharing Web Application (*CookShare*)**. By uniting a modern, reactive user interface (Vue 3, Vite, Pinia) with an enterprise-grade backend architecture (NestJS 11, Prisma, PostgreSQL), the project fulfills all individual project criteria mandated by the Department of Information and Communication Engineering at ITC.

Github:

- **Frontend:** https://github.com/KhornVictor/ip1-e20230078/tree/frontend
- **Backend:** https://github.com/KhornVictor/ip1-e20230078/tree/backend